Can Chance Explain the Mean Difference?

AP Statistics · Topic 7.1 (CED 4.2) · B: 2027-01-12 · E: 2027-02-26 · TEACHER KEY

CED TETHER

- **VAR-1.I** Identify questions suggested by probabilities of errors in statistical inference. [**Skill 1.A**]
- **VAR-1.I.1** Random variation may result in errors in statistical inference.

Essential question: How can rerandomization distinguish a treatment explanation from chance variation while preserving the possibility of inferential error?

DOK-3 skill: 4.B · **FRQ pattern:** `rerandomization-probability-error-and-scope`

DOK rationale: Part (c) adjudicates competing causal claims by coordinating the observed mean difference, a rerandomization tail proportion, its conditional meaning, the remaining chance-error risk, and the distinct roles of random assignment and volunteer recruitment.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK:** 1 → 3 **Minutes:** 45

Teacher does

- Board slide up at the bell. Ask students to decide what 0.009 is a probability of before naming a preferred claim.
- Begin the 7.1 video follow-along at minute 5.
- Return to the board slide at minute 33. Circulate and require students to name the assumption before interpreting the simulated proportion.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Choose Nia or Owen and cite the observed difference or simulation count.
- Complete the follow-along about evidence, chance variation, and rerandomization for a difference in means.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- Where should rerandomized differences be centered if the prompt has no effect?
- Why do we count differences at or below negative 3.6 rather than merely equal to it?
- What appears before and after the conditioning bar in the probability statement?
- Which study feature supports causation, and which missing feature limits generalization?

Adult role

Solo teacher. Circulate 5 pairs. Require separate statements about evidence, error risk, causation, and population scope.

[WARN] WATCH FOR

- Calling 0.009 the probability that the prompt has no effect.
- Using 99.1 percent as certainty that the treatment works.
- Treating random assignment of volunteers as a random sample from the district.

SCORING PART (c) — E / P / I

Expected elements

- **selects-conditional-conclusion** (required) — Selects Nia and links the small tail proportion to weak chance evidence
- **uses-specific-evidence** (required) — Uses 0, -3.6, and 18 of 2,000
- **distinguishes-conditional-directions** (required) — Distinguishes the tail probability under no effect from its reverse
- **bounds-scope** (required) — Uses assignment for causation and volunteers to limit scope
- **recognizes-remaining-error** (required) — Recognizes that a false effect claim remains possible

E	Chooses Nia, conditions 0.009 correctly, uses all anchors, bounds scope, and names remaining error.
P	Supports Nia but incompletely explains conditioning, scope, or remaining error.
I	Calls 0.009 the probability of no effect, claims certainty, or generalizes to the district.

[STAR] TODAY'S PROBLEM — annotated

Suppose a researcher recruits 50 student volunteers to test whether a planning prompt lowers mean logic-task time, then randomly assigns 25 to the prompt and 25 to standard directions. Their means are 14.2 and 17.8 minutes, so prompt minus standard is -3.6. Under no effect, 18 of 2,000 rerandomized differences are -3.6 or lower.

Nia: "A result this favorable is rare under no effect, supporting a causal decrease for students like these volunteers."

Owen: "The 0.009 is the probability of no effect, so there is a 99.1% chance the prompt works for every district student."

Experiment results

Group	n	Mean (min)
Prompt	25	14.2
Standard	25	17.8

FIRST TAKE (5 min)

Whose use of the rerandomization result is better supported, Nia's or Owen's? Cite -3.6 or 18/2,000.
Key: Not graded. Owen's conditional-probability reversal is familiar and tempting; the finish should preserve both strong evidence and a nonzero chance of error.

Rules callout "ASK WHAT CHANCE WOULD DO UNDER NO EFFECT (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** State the directional claim, the observed evidence statistic, and its expected value under no effect.

Key: The claim is that the prompt lowers mean completion time. The observed prompt-minus-standard difference is $14.2 - 17.8 = -3.6$ minutes; under no effect its expected value is 0.

DOK 2 **(b)** Describe one rerandomization trial, compute the tail proportion, and interpret it under no effect.

Key: Keep the 50 times fixed, randomly split them into groups of 25, and record prompt minus standard. Repeating this under no effect gives $18/2{,}000 = 0.009$. Thus, if the prompt has no effect, about 0.9% of assignments produce a difference of -3.6 or lower by chance.

DOK 3 **(c)** Decide whose conclusion is warranted using 0, -3.6, and 18 of 2,000. Distinguish the two directions of conditional probability, bound causation and generalization from the design, identify the remaining inference error, and explain the rejected statement's consequence.

Key: Nia's bounded conclusion is warranted. Under no effect the distribution centers at 0, and only 18 of 2,000 trials reached -3.6 or lower, so 0.009 estimates $P(\text{difference} \leq -3.6 \mid \text{no effect})$, not the reverse conditional probability. Chance is an unconvincing but still possible explanation, so a false effect claim remains possible. Assignment supports causation for these volunteers; volunteer recruitment blocks district-wide generalization. Owen reverses the condition, converts evidence into certainty, and overstates population scope.

EXIT

One thing the video changed about my first take: ______________________________